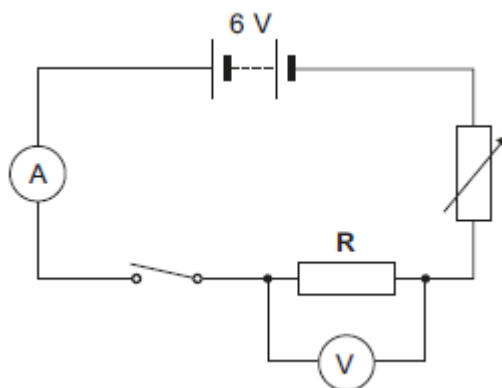


Q1.

The diagram shows an electrical circuit.



- (a) The 6 V battery shown in the diagram is made up of a number of identical 1.5 V cells.

Calculate the minimum number of cells needed to make the battery.

Number of cells = _____

(1)

- (b) The switch in the diagram is shown in the open position. Closing the switch completes the circuit.

Charge flows through the completed circuit and a reading is shown on both the ammeter and the voltmeter.

- (i) In 10 seconds, 20 coulombs of charge flows through the circuit.

Calculate the current reading shown on the ammeter.

Current = _____ A

(2)

- (ii) For 20 coulombs of charge to flow through the resistor R, 100 joules of work must be done.

Calculate the potential difference reading given by the voltmeter.

Potential difference = _____ V

(2)

(Total 5 marks)

Q2.

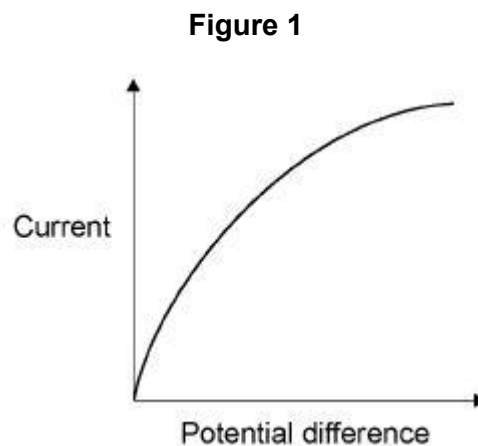
- (a) Complete the sentence. Choose answers from the box.

charge	potential difference	power	temperature	time
---------------	---------------------------------	--------------	--------------------	-------------

The current through an ohmic conductor is directly proportional to the _____ across the component, provided that the _____ remains constant.

(2)

- (b) **Figure 1** shows a current – potential difference graph for a filament lamp.



Explain how the resistance of a filament lamp changes as the potential difference across it increases.

(3)

- (c) Many householders are replacing their filament lamps with LED lamps which are more energy efficient.

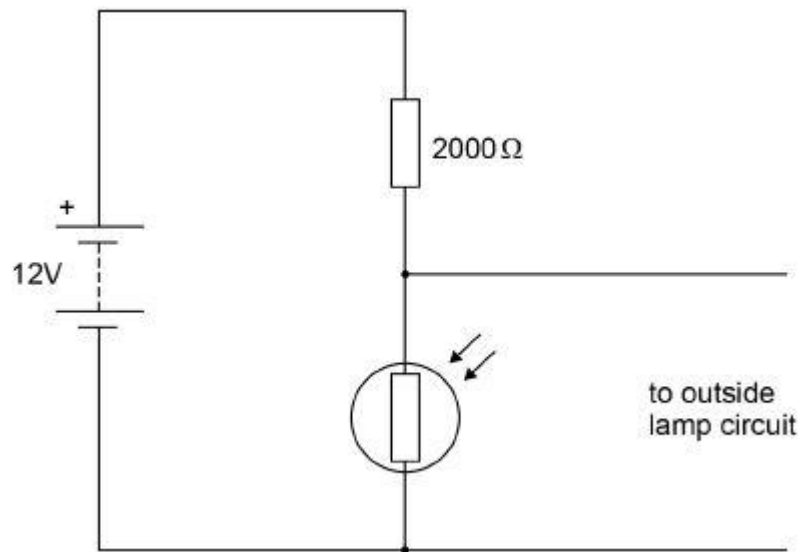
What does more energy efficient mean?

(1)

A Light Dependent Resistor (LDR) is used to turn on an outside lamp when it gets dark.

Part of the circuit is shown in **Figure 2**.

Figure 2



- (d) The light intensity decreases.

What happens to the potential difference across the LDR and the current in the LDR?

Potential difference _____

Current _____

(2)

- (e) What is the resistance of the LDR when the potential difference across it is 4 V?

Give a reason for your answer.

Explain your answer.

Resistance = _____ Ω

Reason _____

(2)

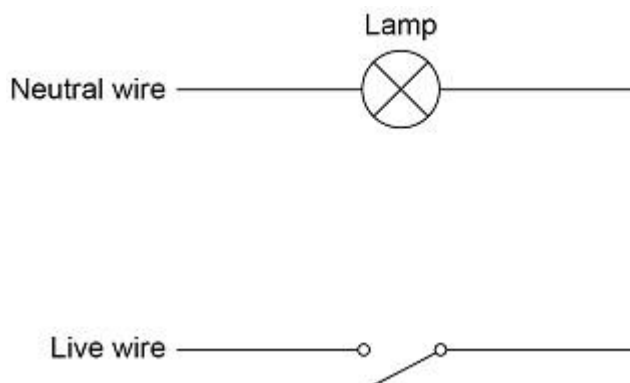
- (f) Calculate the current through the LDR when the resistance of the LDR is $5000\ \Omega$.

Give your answer to 2 significant figures.

Current = _____ A
(4)
(Total 14 marks)

Q3.

The diagram below shows part of a mains electricity lighting circuit in a house.



- (a) A fault in the switch caused a householder to receive a mild electric shock before a safety device switched the circuit off.

The mean power transfer to the person was 5.75 W.

The potential difference across the person was 230 V.

Calculate the resistance of the person.

Resistance = _____ Ω
(5)

- (b) An electrician replaced the switch.

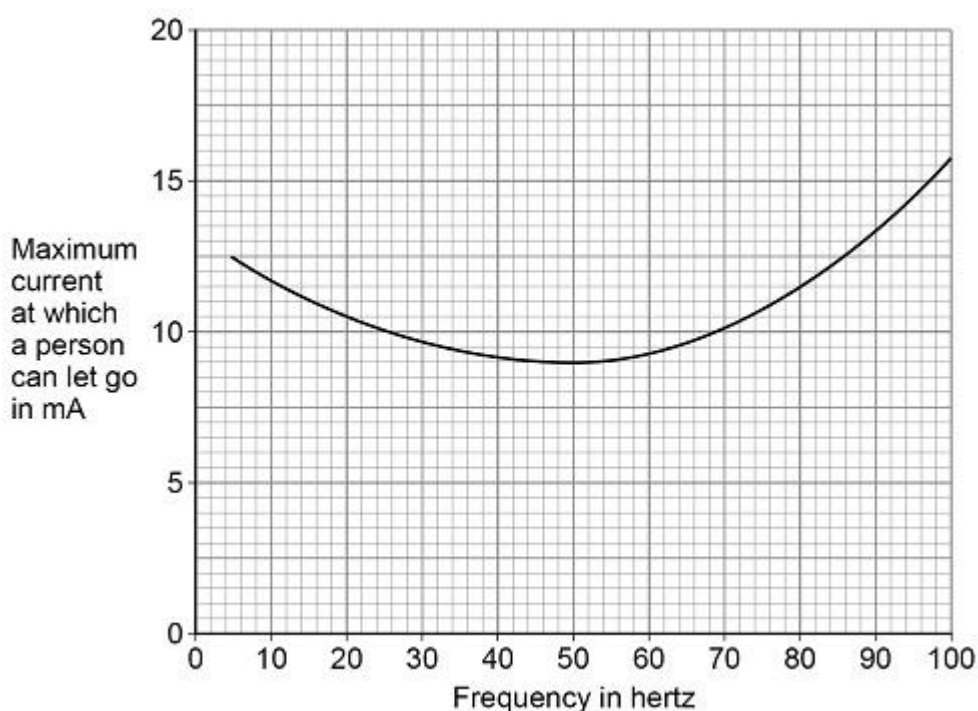
The electrician would have received an electric shock unless the circuit was disconnected from the mains supply.

Explain why.

(3)

- (c) The current from an electric shock causes a person's muscles to contract. The person cannot let go of the electrical circuit if the current is too high.

The graph below shows how the maximum current at which a person can let go depends on the frequency of the electricity supply.



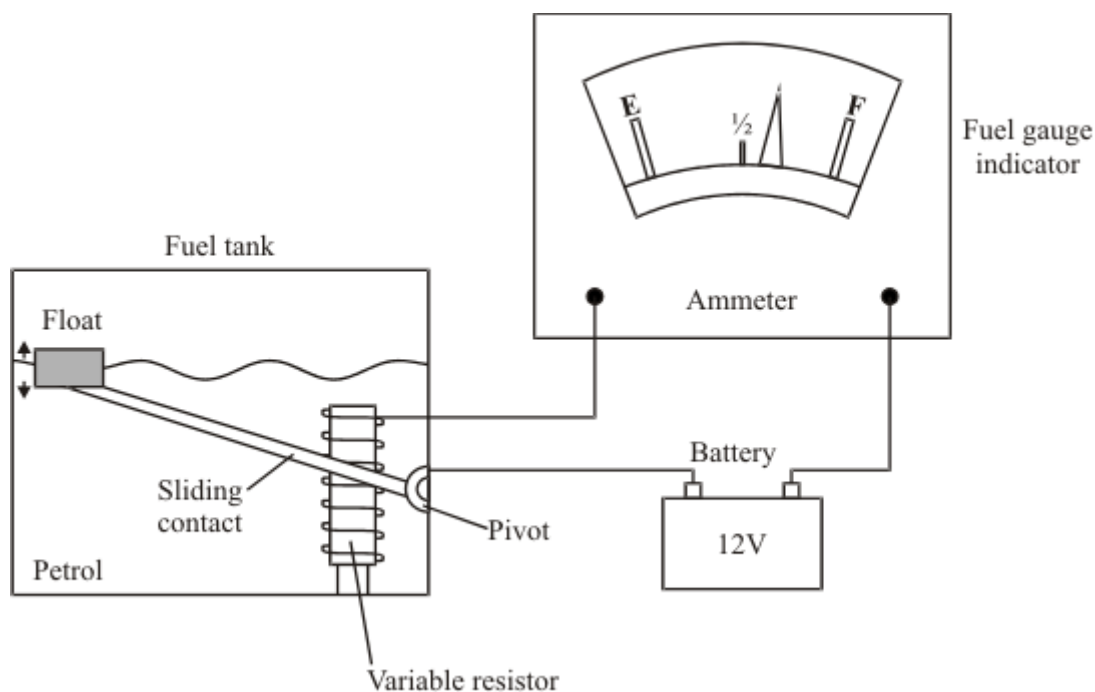
The UK mains frequency is 50 Hz.

Explain why it would be safer if the UK mains frequency was **not** 50 Hz.

Q4.

The diagram shows the fuel gauge assembly in a car.

- The sliding contact touches a coil of wire and moves over it.
- The sliding contact and the coil form a variable resistor.
- The sliding contact is connected to a float via a pivot.
- The fuel gauge indicator is an ammeter.
- When the petrol level changes, the resistance of the circuit changes.
- This causes the pointer in the fuel gauge indicator to move.



- (a) Use standard symbols to draw a circuit diagram for the fuel gauge assembly.

(3)

- (b) How will the current in the circuit change as the level of petrol in the tank falls?

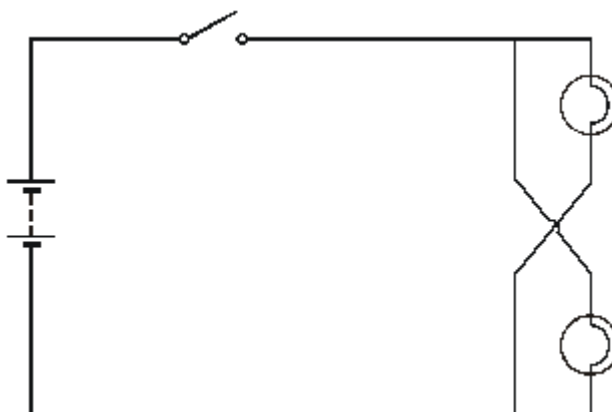
Explain the reason for your answer.

(2)

(Total 5 marks)

Q5.

The circuit diagram below shows a circuit used to supply electrical energy to the two headlights of a car.



The current through the filament of one car headlight is 3.0 A. The potential difference across each of the two headlights is 12 V.

- (a) Suggest a suitable fuse for the circuit. _____

(1)

- (b) Calculate the resistance of the headlight filament when in use.

Answer _____ W

(2)

- (c) Calculate the power supplied to the two headlights of the car.

Answer _____ W

(2)

- (d) The fully charged car battery can deliver 72 kJ of energy at 12 V. How long can the battery keep the headlights fully on?

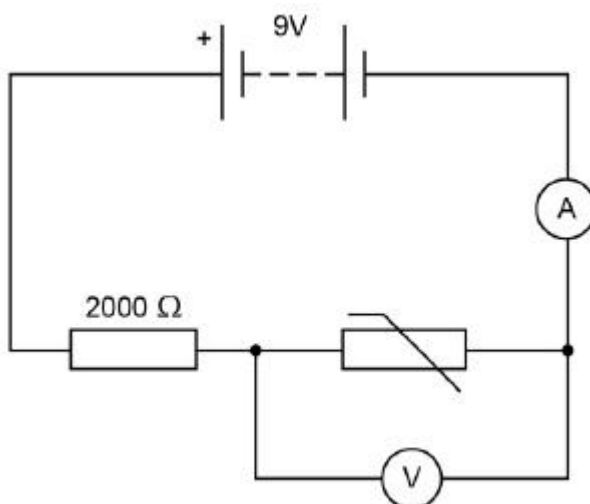
Answer _____ s

(2)

(Total 7 marks)

Q6.

The diagram shows a temperature sensing circuit used to control a heating system in a house.



- (a) What quantity does the ammeter measure?

(1)

- (b) The current in the circuit is 3.5 mA when the potential difference across the thermistor is 4.2 V

Calculate the resistance of the thermistor.

Resistance = _____ Ω

(3)

- (c) Calculate the charge that flows through the thermistor in 5 minutes when the current is 3.5 mA.

Charge = _____ C

(3)

- (d) Explain why the potential difference across the thermistor changes as the temperature in the house decreases.

(2)

- (e) The circuit shown in the diagram can be modified to turn lights on and off by replacing the thermistor with a Light Dependent Resistor (LDR).

Draw the circuit symbol for an LDR in the space below.

(1)

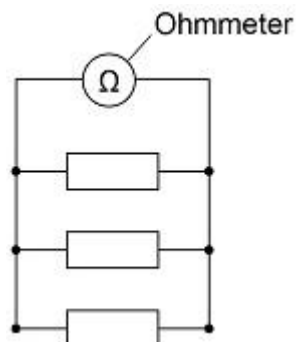
(Total 10 marks)

Q7.

A student investigated how the total resistance of identical resistors connected in parallel varied with the number of resistors.

The student used an ohmmeter to measure the total resistance of the resistors.

The diagram below shows the student's circuit with 3 resistors.



The student repeated each reading of resistance three times.

The table below shows some of the results for 3 resistors in parallel.

Number of resistors	Total resistance in ohms			
	Reading 1	Reading 2	Reading 3	Mean
3	15.8	15.3	X	15.7

- (a) Calculate value **X** in the table above.

$$X = \underline{\hspace{2cm}} \Omega$$

(2)

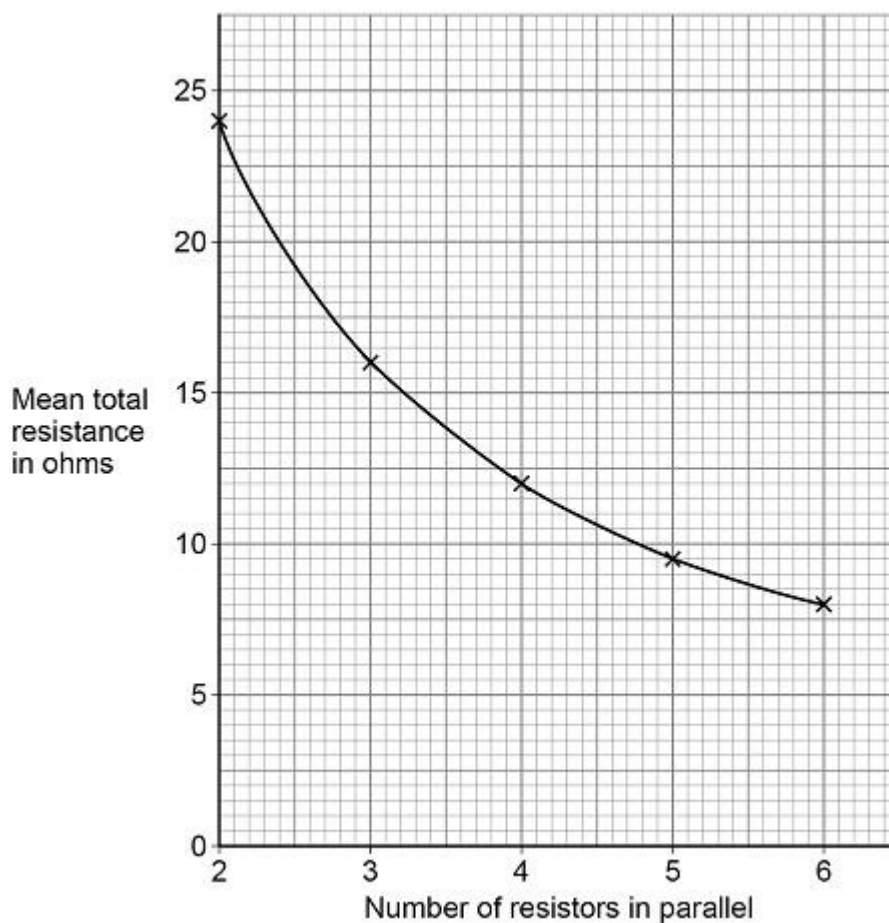
- (b) The student thought that taking a fourth reading would improve the precision of the results.

The fourth reading was 16.2Ω .

Explain why the student was wrong.

(2)

The graph below shows the results from the investigation.



- (c) The student concluded that the number of resistors in parallel was inversely proportional to the mean total resistance.

Explain why the student was correct.

Use data from the graph in your answer.

(3)

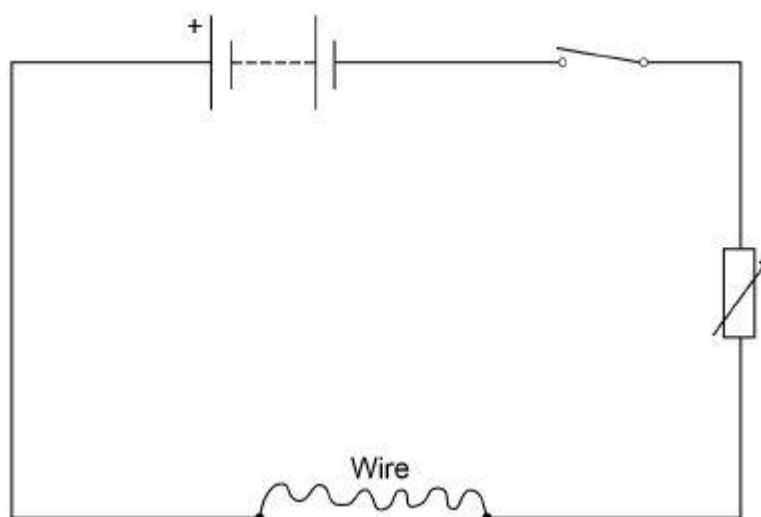
- (d) Explain why adding resistors in parallel decreases the total resistance.

Q8.

A student investigated how the resistance of a piece of nichrome wire varies with length.

Figure 1 shows part of the circuit the student used.

Figure 1



- (a) Complete **Figure 1** by adding an ammeter and a voltmeter.

Use the correct circuit symbols.

(3)

- (b) Describe how the student would obtain the data needed for the investigation.

Your answer should include a risk assessment for **one** hazard in the investigation.

(6)

- (c) Why would switching off the circuit between readings have improved the accuracy of the student's investigation?

Tick **one** box.

The charge flow through the wire would not change.

☐

The potential difference of the battery would not increase.

☐

The power output of the battery would not increase.

☐

The temperature of the wire would not change.

☐

(1)

- (d) The student used crocodile clips to make connections to the wire.

They could have used a piece of equipment called a 'jockey'.

Figure 2 shows a crocodile clip and a jockey in contact with a wire.

Figure 2



Crocodile clip



Jockey

How would using the jockey have affected the accuracy and resolution of the student's results compared to using the crocodile clip?

Tick **two** boxes.

The accuracy of the student's results would be higher.

☐

The accuracy of the student's results would be lower.

☐

The accuracy of the student's results would be the same.

☐

The resolution of the length measurement would be higher.

☐

The resolution of the length measurement would be lower.

☐

The resolution of the length measurement would be the same.

☐

(2)
(Total 12 marks)

Mark schemes

Q1.

- (a) 4 1
- (b) (i) 2
- allow 1 mark for correct substitution ie*
- $I = \frac{100}{20}$
- provided no subsequent step* 2
- (ii) 5
- allow 1 mark for correct substitution ie*
- $V = \frac{100}{20}$
- provided no subsequent step* 2
- [5]**

Q2.

- (a) potential difference 1
- allow p.d.*
- allow voltage*
- temperature 1
- in this order only*
- (b) the current increases (when the potential difference increases) 1
- (which) causes the temperature of the filament to increase 1
- (so) the resistance increases
- do **not** accept resistance increases and then levels off* 1
- (c) a higher proportion / percentage of the (total) power / energy input is usefully transferred
- wastes less energy is insufficient*
- or**
- higher (useful) power / energy output for the same (total) power / energy input 1
- (d) potential difference increases 1
- current decreases

1

(e) $1000\ (\Omega)$

reason only scores if $R = 1000\ (\Omega)$

1

potential difference is shared in proportion to the resistance

allow a justification using a correct calculation

1

(f) $12 = I \times 7000$

1

$$I = \frac{12}{7000}$$

1

$$I = 1.71 \times 10^{-3}\ (\text{A})$$

an answer that rounds to $1.7 \times 10^{-3}\ (\text{A})$ scores 3 marks

1

$$I = 1.7 \times 10^{-3}\ (\text{A})$$

this answer only

or

$$I = 0.0017\ (\text{A})$$

an answer of $2.4 \times 10^{-3}\ (\text{A})$ scores 2 marks

if no other marks scored allow 1 mark for calculation of total resistance ($7000\ \Omega$)

1

an answer of $1.7 \times 10^{-3}\ (\text{A})$ scores 4 marks

[14]

Q3.

(a) $5.75 = I \times 230$

1

$$I = \frac{5.75}{230}$$

1

$$I = 0.025\ (\text{A})$$

1

$$230 = 0.025 \times R$$

or

$$R = \frac{230}{0.025}$$

allow a correct substitution using an incorrect value of I

or

allow a correct rearrangement using incorrect value of I

1

$$R = 9200 (\Omega)$$

allow a correct calculation of resistance using an incorrect value of I

alternative approach for 4th and 5th marks:

$$5.75 = 0.025^2 \times R \quad (1)$$

or

$$R = \frac{5.75}{0.025^2}$$

$$R = 9200 (\Omega) \quad (1)$$

alternative approach:

$$5.75 = \frac{230^2}{R} \quad (3)$$

$$R = 9200 (\Omega) \quad (1)$$

1

- (b) one wire in the switch is live

allow the switch / circuit is live allow one wire is at a potential of 230 V

1

the electrician is earthed

or

the electrician is at earth potential

1

(so) there will be a (large) potential difference between the live wire and the electrician / earth (if the electrician touched the wire)

1

- (c) 50 Hz has the lowest (maximum) let-go current

1

a higher / lower / different frequency would allow people to let go at a greater current

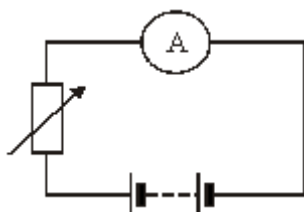
allow a specific numerical example as opposed to a trend

1

[10]

Q4.

- (a)



3

one mark for each symbol
allow more than 2 cells joined

max. 2 marks if symbols incorrectly allow rheostat arrow in

either direction

- (b) current will decrease

1

since resistance greater

1

[5]

Q5.

- (a) in range $6 < I \leq 13$ A

for 1 mark

(no unit no mark)

1

- (b) 4

gains 2 marks

(else working

gains 1 mark

(resistance of circuit correctly worked (2Ω))

2

- (c) $72 (I^2 R)$ ecf

gains 2 marks

else working

gains 1 mark

an answer of 36W (ie for one lamp) – (1)

2

- (d) 1000 or 16.7 min (ecf from (c))

gains 2 marks

else working

gains 1 mark

(formula with incorrect substitution – no mark (12V))

2

[7]

Q6.

- (a) current

1

- (b) $4.2 = 3.5 \times 10^{-3} \times R$

1

$$R = 4.2 / 3.5 \times 10^{-3}$$

1

$$R = 1200 (\Omega)$$

an answer of 1200 (Ω) scores 3 marks

an answer of 1.2 scores 2 marks

- (c) conversion from minutes to seconds (300 s)

$$Q = 0.0035 \times (5 \times 60)$$

$$Q = 1.05 \text{ C}$$

an answer of 1.05 (C) scores 3 marks

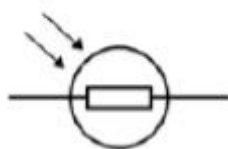
an answer of 17.5 scores 1 mark

an answer of 1050 or 0.0175 scores 2 marks

- (d) (potential difference) increases

(because thermistor) resistance increases

2nd mark dependent on scoring 1st mark



- (e)

[10]

Q7.

(a) $15.7 = \frac{15.8 + 15.3 + X}{3}$

$$X = 16.0 \text{ (}\Omega\text{)}$$

- (b) precise results show little variation

the 4th result was further away from the mean than the other values

allow the range of values has increased

ignore the 4th result was an anomaly

- (c) two pairs of values of n and R showing that $n \times R = \text{constant}$

e.g. $2 \times 24 = 48$, $3 \times 16 = 48$

$4 \times 12 = 48$, $5 \times 9.5 = 47.5$

$6 \times 8 = 48$

third pair of values of n and R showing that $n \times R = \text{constant}$

(so) $n \times R = \text{constant}$ (showing the student was correct)

allow 1 mark each for two statements relating the change in

*number of resistors to the change in (mean total) resistance
allow 1 mark for use of data from graph to confirm at least one
statement*

1

- (d) multiple paths for charge / electrons to flow
allow current for charge

1

total current is greater (for the same potential difference when more resistors are added)

1

[9]

Q8.

- (a) ammeter and voltmeter symbols correct

1

voltmeter in parallel with wire

1

ammeter in series with wire

1

- (b) **Level 3:** The method would lead to the production of a valid outcome. All key steps are identified and logically sequenced.

5-6

Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the method is not fully logically sequenced.

3-4

Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.

1-2

No relevant content

0

Indicative content

- length measured
- length varied
- current measured
- potential difference measured
- repeat readings
- calculate resistance for each length
- $\text{resistance} = \frac{\text{potential difference}}{\text{current}}$
- plot a graph of resistance against length
- hazard: high current
- may cause wire to melt / overheat
- may cause burns (to skin)
- use low currents

- (c) the temperature of the wire would not change

1

(d) the accuracy of the student's results would be higher

1

the resolution of the length measurement would be higher

1

[12]

Examiner reports

Q1.

- (a) Just over half of the students scored this mark. A common incorrect answer was 9, arrived at by multiplying 6 by 1.5.
- (b)
 - (i) This question was answered well with the majority of the students scoring both marks. The most usual incorrect answer was where students had multiplied the two numerical values instead of dividing them.
 - (ii) Just over half of the students scored these two marks. The most common errors were multiplying the work done by the charge flowing or dividing the charge by the work done.

Q2.

- (a) 57% of students scored 2 marks for this question, with 32% scoring 1 mark. The most common mistake was not stating which quantity should remain constant, 'charge' being a commonly seen incorrect answer.
- (b) This question discriminated well between students with 13% of students scoring 3 marks, 30% scoring 2 marks and 40% scoring 1 mark. Some students incorrectly stated that the current decreased or the resistance decreased. Contradictory statements often negated marks, e.g. pd increases causing current to increase but then decrease.
- (c) 24% of students answered this question correctly. Students needed to answer in terms of amount / proportion / percentage input compared to useful output in order to score the mark.
- (d) This question differentiated well, with 26% of students scoring 2 marks, while 39% of students scored 1 mark.
- (e) 13% of students scored 2 marks on this question. Very few correct answers of 1000 ohms were seen. When 1000 ohms was given, the reason usually justified the award of 2 marks, so very few 1 mark answers were seen.
- (f) 12% of students scored 4 marks on this question. Many students failed to add the resistance of the two series resistors together. The most common incorrect answer was 0.0024 (A), which scored 2 marks. Most students who calculated the correct current also correctly showed their answer to 2 significant figures.

Q3.

- (a) 53% of students scored 5 marks on this multi-step calculation. Lower-scoring students didn't score marks for stating and trying to use incorrect physics equations or rearranging correct equations incorrectly. The main reasons for not scoring full marks included: knowing one form of an equation correctly, but re-arranging it incorrectly, eg stating $P = I V$ then rearranging it to $I = V/P$, mixing up two half-known equations eg $P = I^2 R$, not realising that the 5 marks for this question warranted more than using one basic equation, eg stating $R = V/I$ and substituting the given value of power for I .
- (b) Students often didn't score the third mark because of their technical inaccuracies referring to potential difference of the live wire or the electrician instead of potential.

Many students mixed up quantities like potential difference with current and charge in confused answers. The question discriminated well between students with 15% of students scoring 3 marks, 15% of students scoring 2 marks and 35% of students scoring 1 mark.

- (c) 18% of students scored 2 marks, while 44% of students scored 1 mark. A good proportion of students appeared to understand what the graph was showing, but often had difficulty in expressing their ideas clearly. A number of students thought that a higher current would be a disadvantage, so stated either that the current was lower at frequencies other than 50Hz, or 're-wrote' the question to answer why 50Hz was the safest frequency. Simply reading off values from the graph was insufficient without an explanation of what the implications were.

Q4.

- (a) Hardly any candidates were able to draw a completely correct circuit diagram, although most were able to score one or two marks for two correct symbols. The main stumbling block was the symbol for a variable resistor.
- (b) Although a number were able to state that the current would decrease, few candidates were able to go on and explain why, in terms of the resistance increasing.

Q5.

In (a) the fuse chosen was often too small, the parallel alignment of the lamps being missed. The calculations in (b) and (c) were moderately well done. The time calculation in (d) proved to be far more difficult.

Q7.

- (a) 91% of students correctly calculated the value in the table.
- (b) Only 4% of students scored 2 marks, while 24% of students scored 1 mark. Many students continue to think that precision is related to the resolution of an instrument and score zero. Statements like 'it increases the accuracy not the precision' were common and insufficient to score a mark. Some students thought that the 3 readings were related to the fact that there were 3 resistors, so the 4th reading wasn't needed as there were only 3 resistors. An answer of 'the 4th result increased the mean' was insufficient as the mean would have always increased if the result had been greater than the mean.
- (c) Inverse proportion is a difficult concept for some students - many stated the general trend that as one quantity increases the other quantity decreases, scoring no marks. Only 2% of students examined the relationship in enough detail to score 3 marks. Very few students calculated the constant of proportionality.
- (d) This explanation was not well known, only 11% of students scored 2 marks, while 26% of students scored 1 mark. The most common answer was that 'the resistance is shared'. Another common answer was to use the equation for resistors in parallel as an explanation. Of those students who had the correct idea, most gained a mark for stating that there would be multiple paths for the current to take, but very few were able to progress to the idea that the total current in the circuit would be greater.

Q8.

Foundation

- (a) Many students re-drew the circuit with the addition of the voltmeter and ammeter in the space below the question. Normally, ammeter and voltmeter symbols should not have a line drawn through the centre but this was ignored if students had added them to the existing diagram.
 Nearly three quarters of students were able to score two or three marks. Almost all of the answers showed the correct circuit symbols for the ammeter and voltmeter, although a few had the letters A and V within square boxes. The vast majority were able to position the ammeter correctly in the series circuit. The most common error was to show the voltmeter in series.
- (b) Most students had some idea of how to perform this investigation, but generally did not include the details which would lead to a valid outcome. The three quantities that needed to be measured to ensure a valid outcome were length, potential difference and current. In addition to this, to reach Level 3, students needed to state how the resistance would be calculated and include an appropriate hazard. Just over 2% of students matched these criteria to score five or six marks.
 The criteria for Level 2 still required the method to lead to a valid outcome, so measurement of length, potential difference and current were still needed. A further 18% of students fell into this category, scoring three or four marks.
 Students who did not meet these criteria were able to score one or two marks for making some relevant points. A common response was 'measure its resistance' without giving details of how this would be done. Other students described taking ammeter and voltmeter readings, without referring to current and potential difference, but did not mention changing the length.
 Many realised that the wire could get hot, which would be a hazard, but there were a number of responses which referred to 'electrocution' and 'electric shock'; this was judged to be insufficient for a hazard as low voltage supplies are likely to be used.
- (c) Nearly 40% of students gave the correct answer that the temperature of the wire would not change.
- (d) Around one third of students scored both marks, with a further 45% scoring one mark.

Higher

- (a) 65% of students scored 3 marks. A very common mistake was to switch the positions of the ammeter and voltmeter or put them both in series. A large number redrew the circuit in the space below Figure 1. In this case, students needed to indicate which circuit was the one to mark, otherwise the list principle applied.
- (b) 56% of students scored 4 or more marks in this question. The question rewarded students who had clearly prepared methods for the required practical activity. Students had a circuit diagram to refer to, but many did not mention varying the length of the wire, which limited them to Level 1. The three quantities that need to be measured to ensure a valid outcome were length, potential difference and current. In addition to this students needed to state how the resistance would be calculated and include an appropriate hazard to reach Level 3. Risk of electric shock was judged to be insufficient for a hazard as low voltage supplies are likely to be used in schools. Quite a few students stated that the resistance of the wire could be measured using the variable resistor, which limited them to Level 1.
- (c) 73% of students answered this question correctly.
- (d) 43% scored 2 marks on this question, 94% of students scored at least 1 mark.